

## Quiz Question

**Time: 20 minutes**

**Marks: 20**

**Scenario.** Students of **Class A** sometimes attend class mainly to keep their attendance percentage safe. The department designs a simple RNN named **AttendanceSequenceNet** to predict whether a student will attend the next class using previous attendance records.

For each day:

$$x_t = \begin{cases} 1, & \text{student was present} \\ 0, & \text{student was absent} \end{cases}$$

The RNN uses the simplified recurrence and  $h_0 = 0$  where  $W_1$  is the input weight and  $W_2$  is the recurrent weight.

**Given:**

$$W_1 = 1.5, \quad W_2 = 0.5$$

A student has the following attendance sequence for the last three classes:

$$x_1 = 1, \quad x_2 = 0, \quad x_3 = 1$$

1. [4 marks] Compute  $h_1, h_2, h_3$ .
2. [3 marks] Write  $h_3$  directly in terms of  $x_1, x_2, x_3, W_1, W_2$ , without calculating step by step.
3. [3 marks] Which input has the strongest direct contribution to  $h_3$ :  $x_1$ ,  $x_2$ , or  $x_3$ ? Justify using the powers of  $W_2$ .
4. [4 marks] Suppose another student has sequence:

$$x_1 = 1, \quad x_2 = 1, \quad x_3 = 0$$

Compute  $h_3$ . Compare briefly with the first student.

5. [3 marks] If  $W_2 = 0.2$  instead of 0.5, would older attendance records have more or less effect on the final hidden state? Explain briefly.
6. [3 marks] A student claims: "RNN creates new weights for every class day." Is the claim correct? Justify briefly.